Università degli Studi di
Modena e Reggio Emilia
Corso di Laurea in Ingegneria
Informatica

Augmented Reality Framework

AR Foundation

Candidato: Manuel Gherardi
Relatore: Nicola Bicocchi
Fatto con: Canva

Anno Accademico: 2025/2026

UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Cos’è l’AR?

COME VIENE VISUALIZZATA

Spatial AR



See-through AR



Optical See-through



Video See-through (Passthrough)

COME IL DISPOSITIVO RICONOSCE LO SPAZIO O GLI OGGETTI

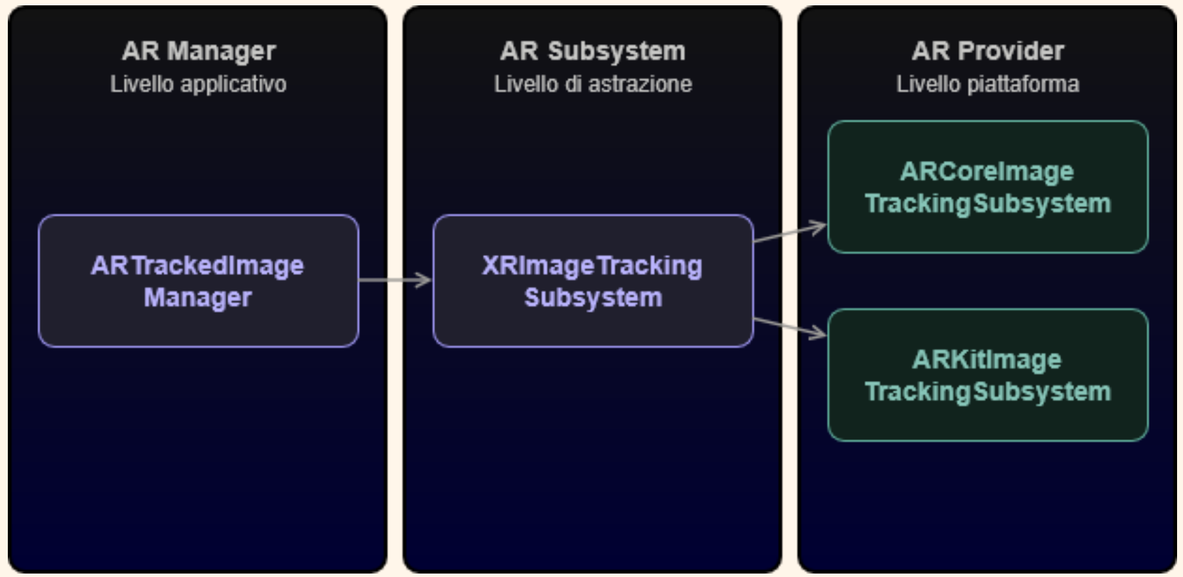
Marker Based vs Markerless

Come Funziona?

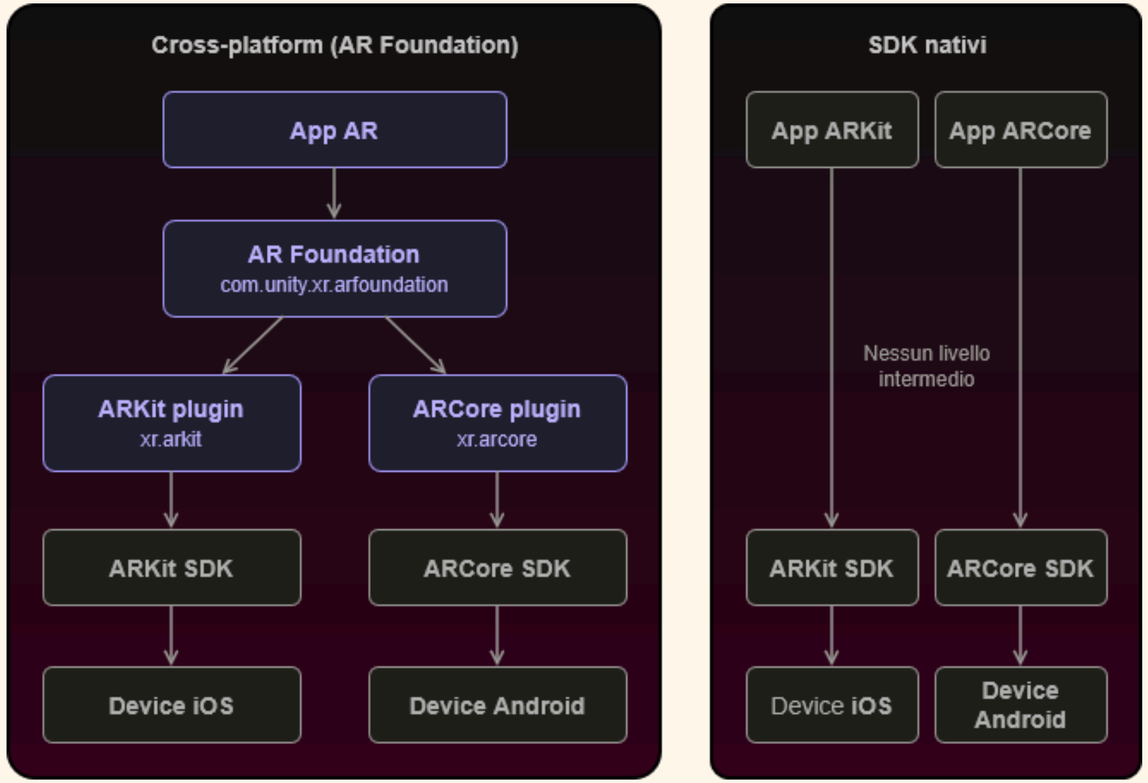


AR Foundation

Architettura a basso livello

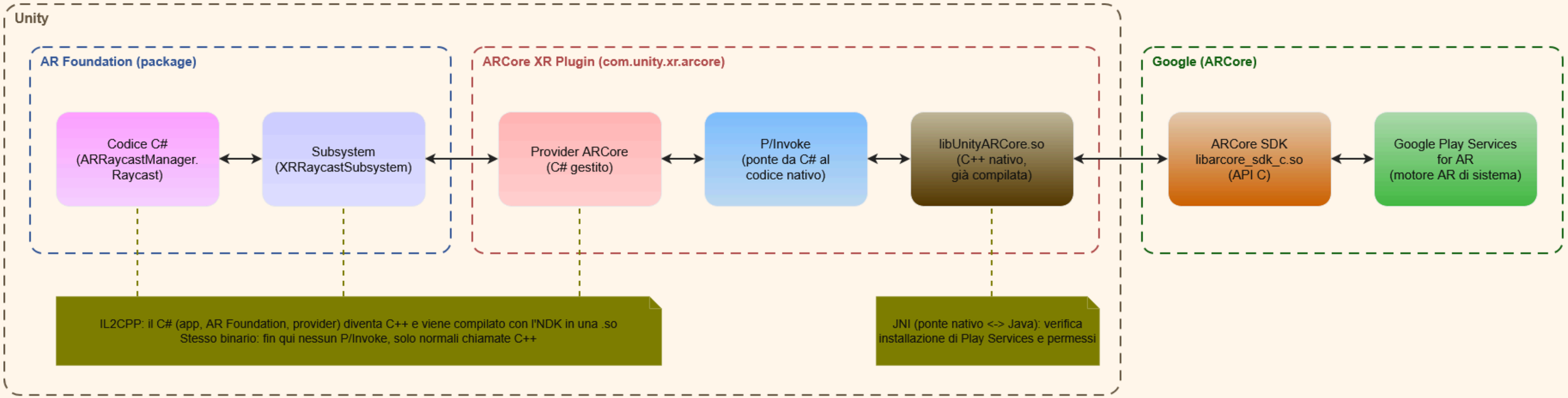


Architettura ad alto livello



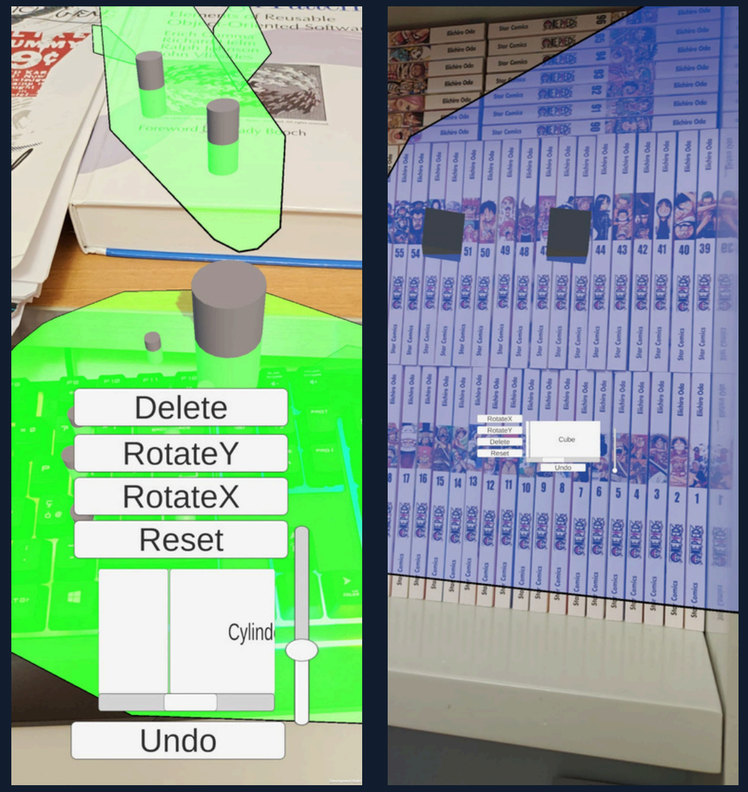
Come funziona a Runtime?

Cosa succede a runtime: da C# al motore AR di sistema

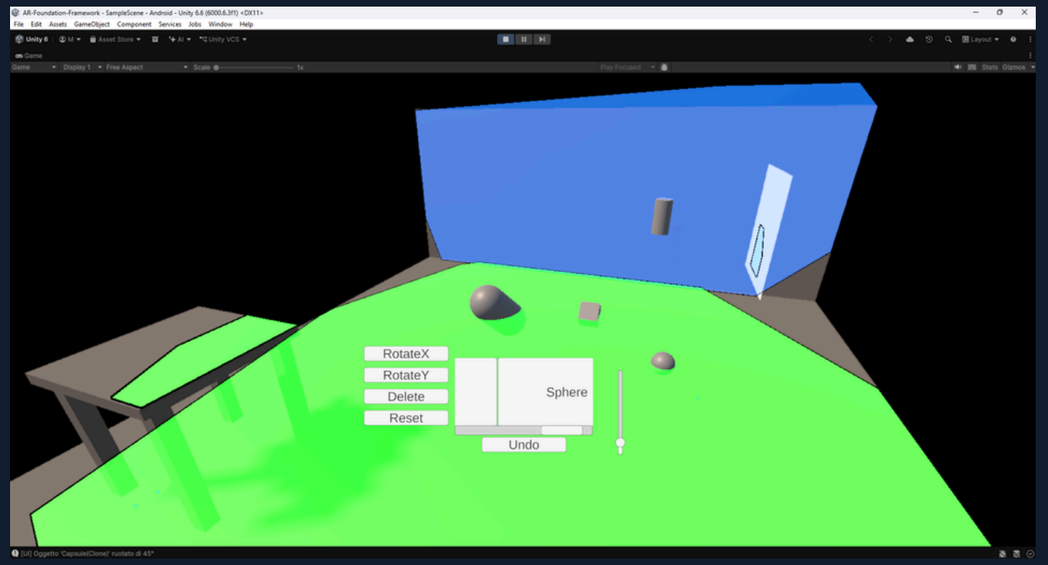


Frecce bidirezionali: i risultati fanno il percorso inverso e arrivano allo sviluppatore (ARPlane, hit del raycast)

App

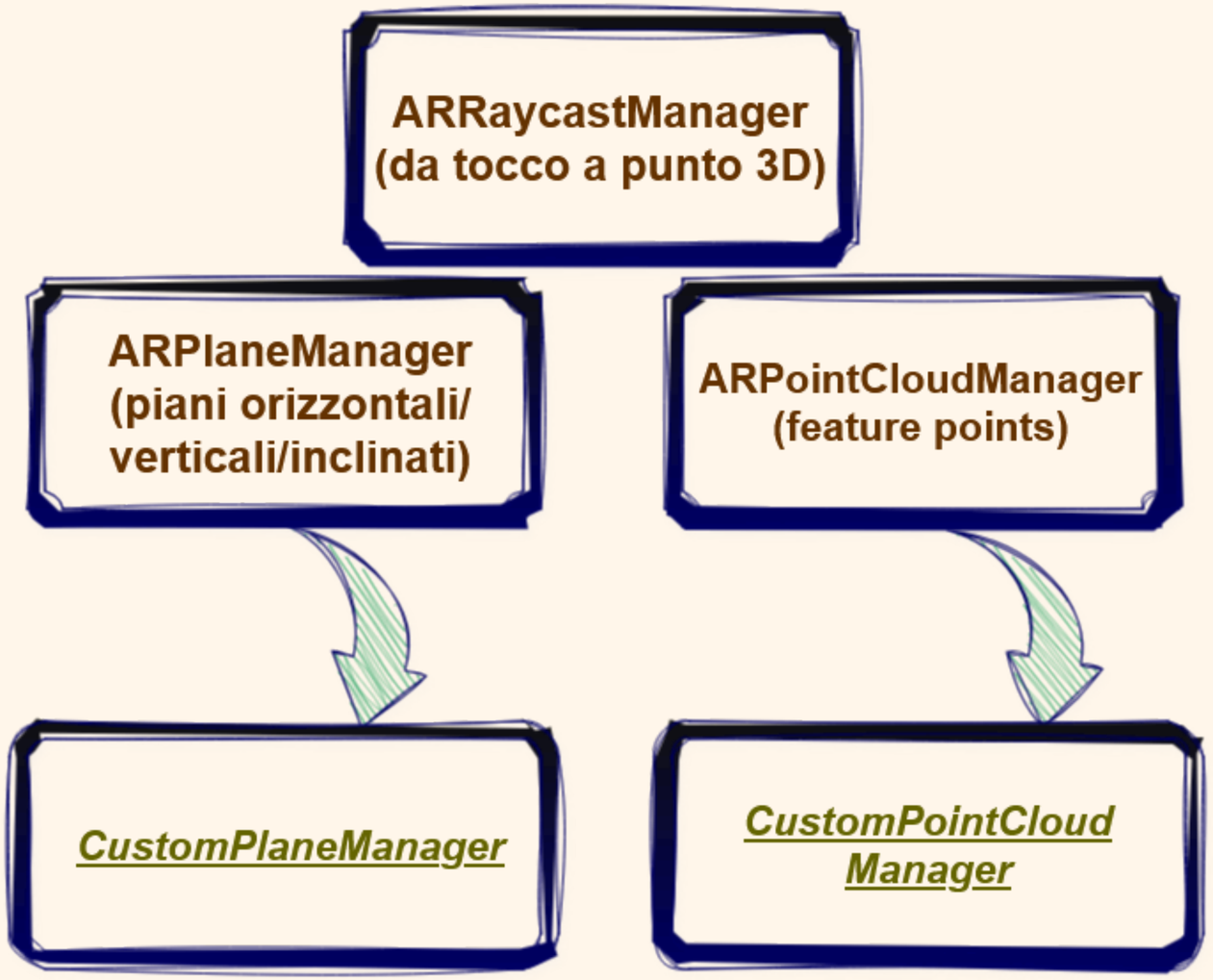


Mobile, in my smartphone

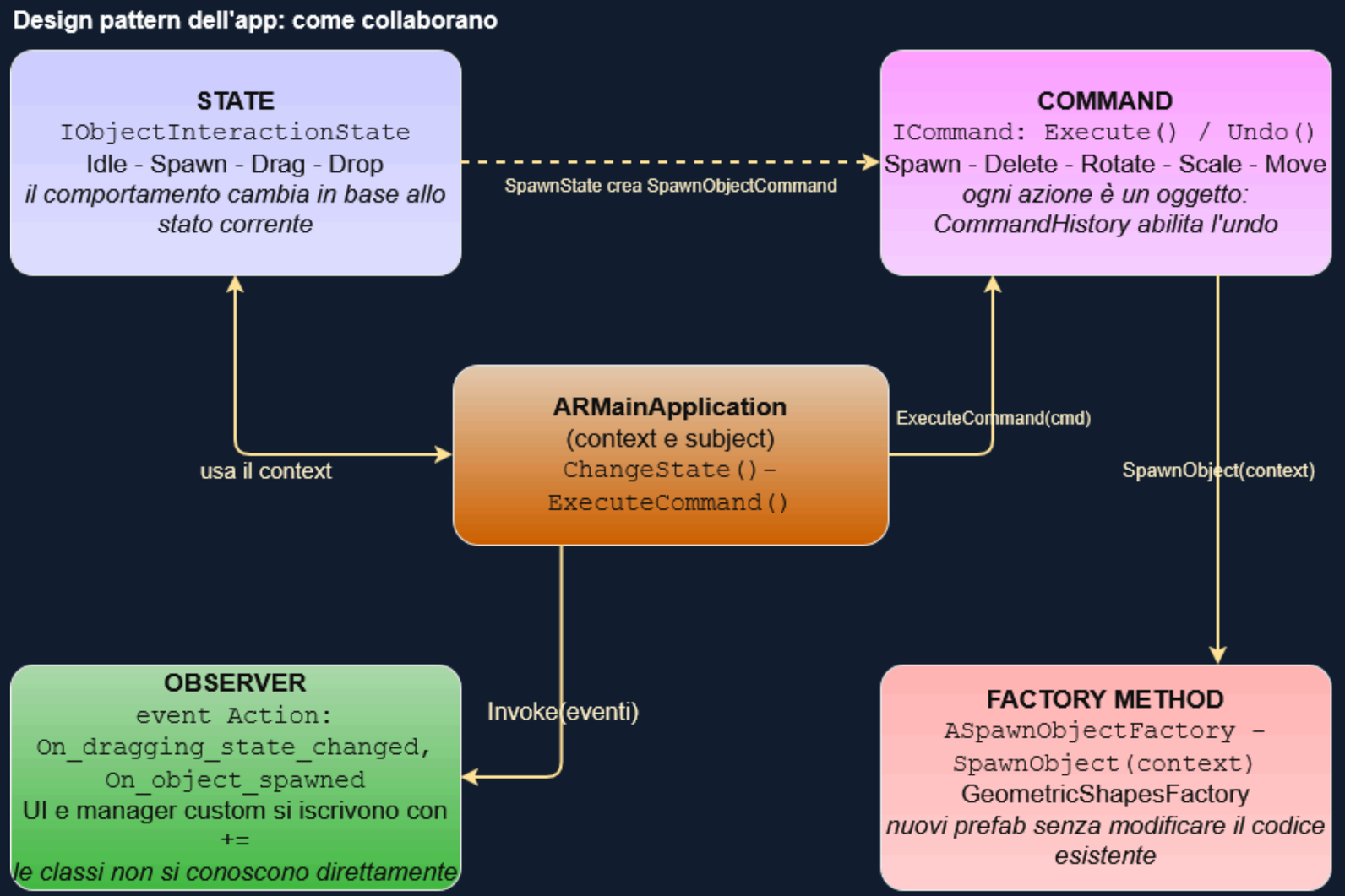


XR Simulator, in the Unity Editor

Componenti AR Usati



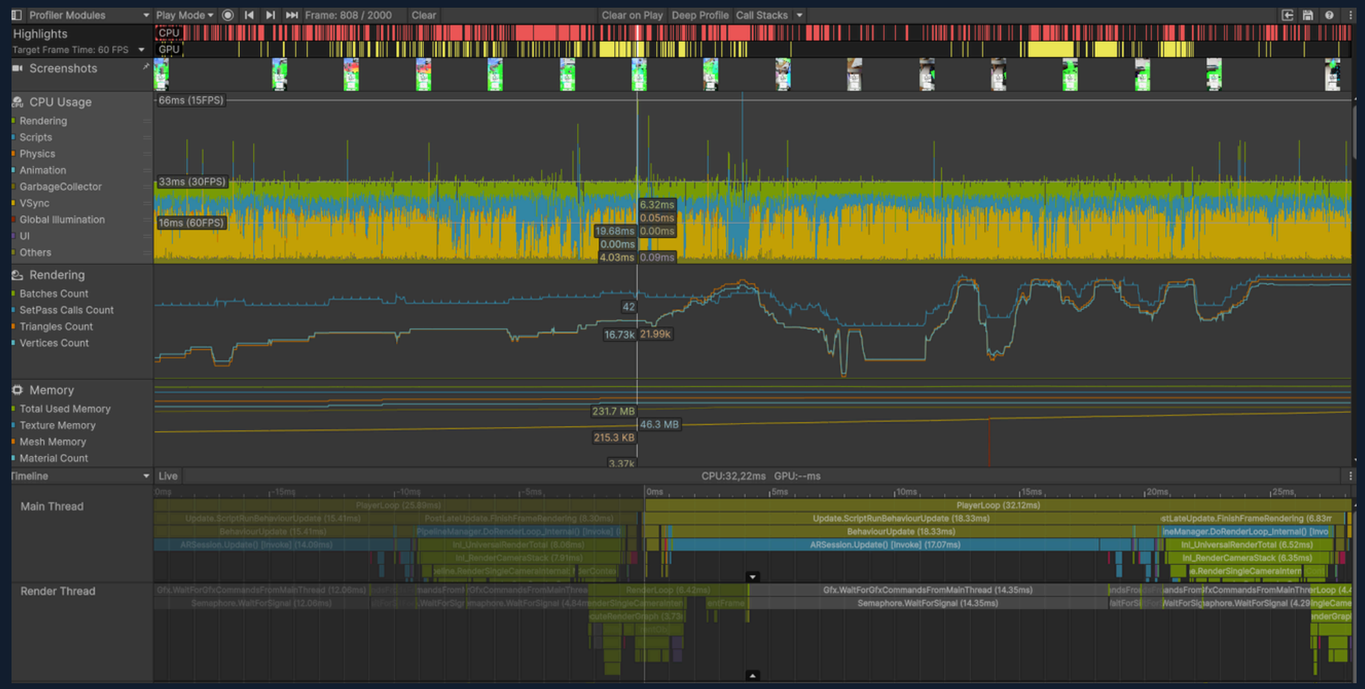
Design Pattern



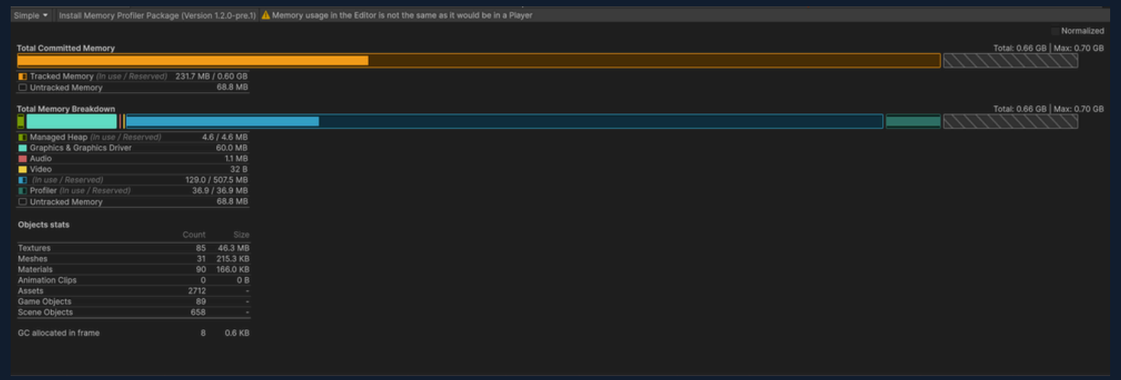
Esempio, spawn di un oggetto: tocco su un piano SpawnState -> SpawnObjectCommand -> Factory crea il prefab -> NotifyObjectSpawned (Observer)

Profiling & Testing: Unity Profiler

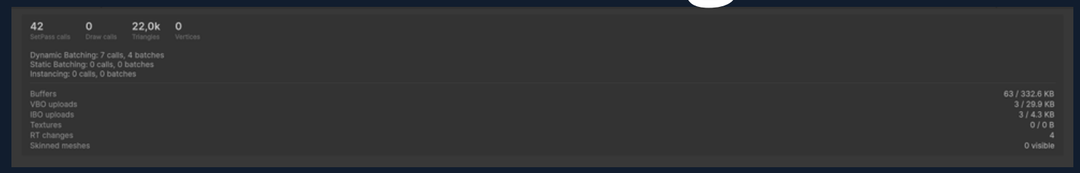
Generale



Memory

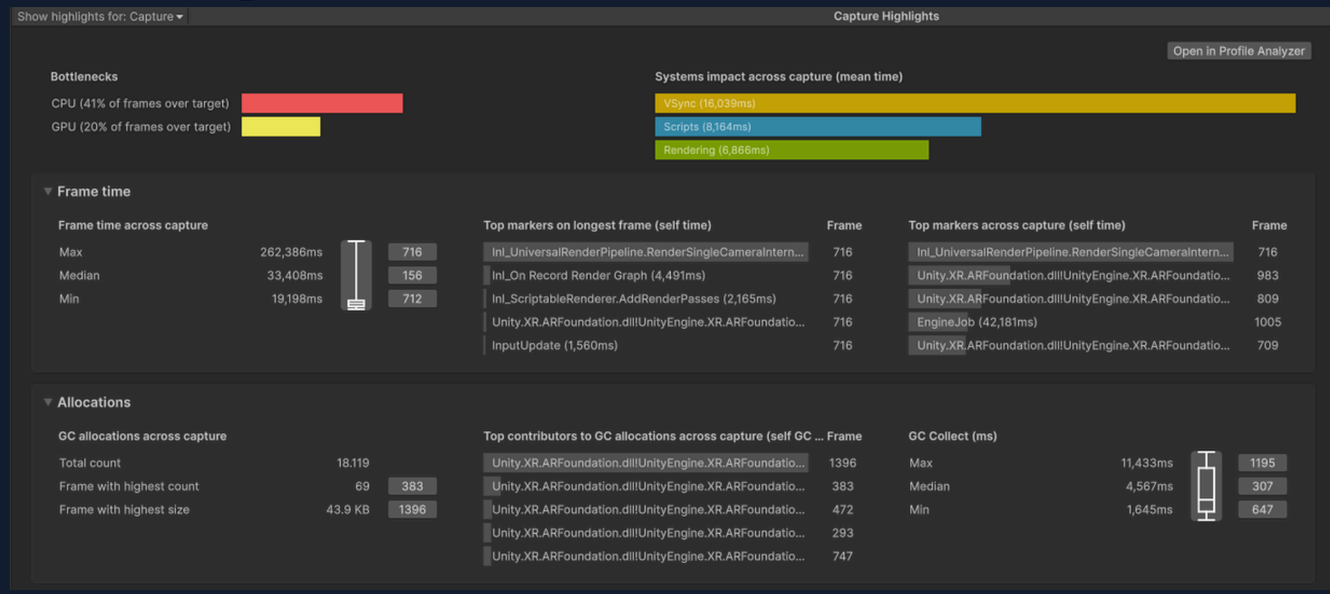


Rendering

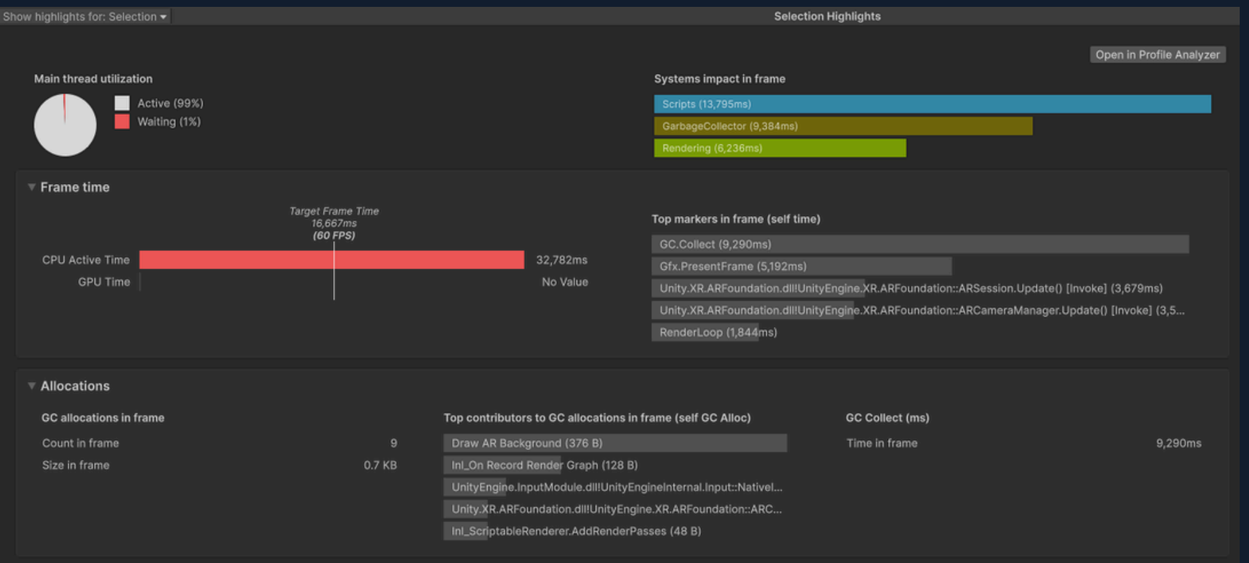


Profiling & Testing

Highlights Capture (Globale)



Highlights Selection (Locale)



Conclusione

ARCHITETTURA

AR Foundation

ARCore / ARKit

PROFILING & TESTING

Performance

Criticità

APPLICAZIONI

Interior Design & E-commerce

Didattica & Architettura

SVILUPPI FUTURI

Ottimizzazione Codice

Configurazione AR & Test

Grazie per l’attenzione